



xDM Developer Training

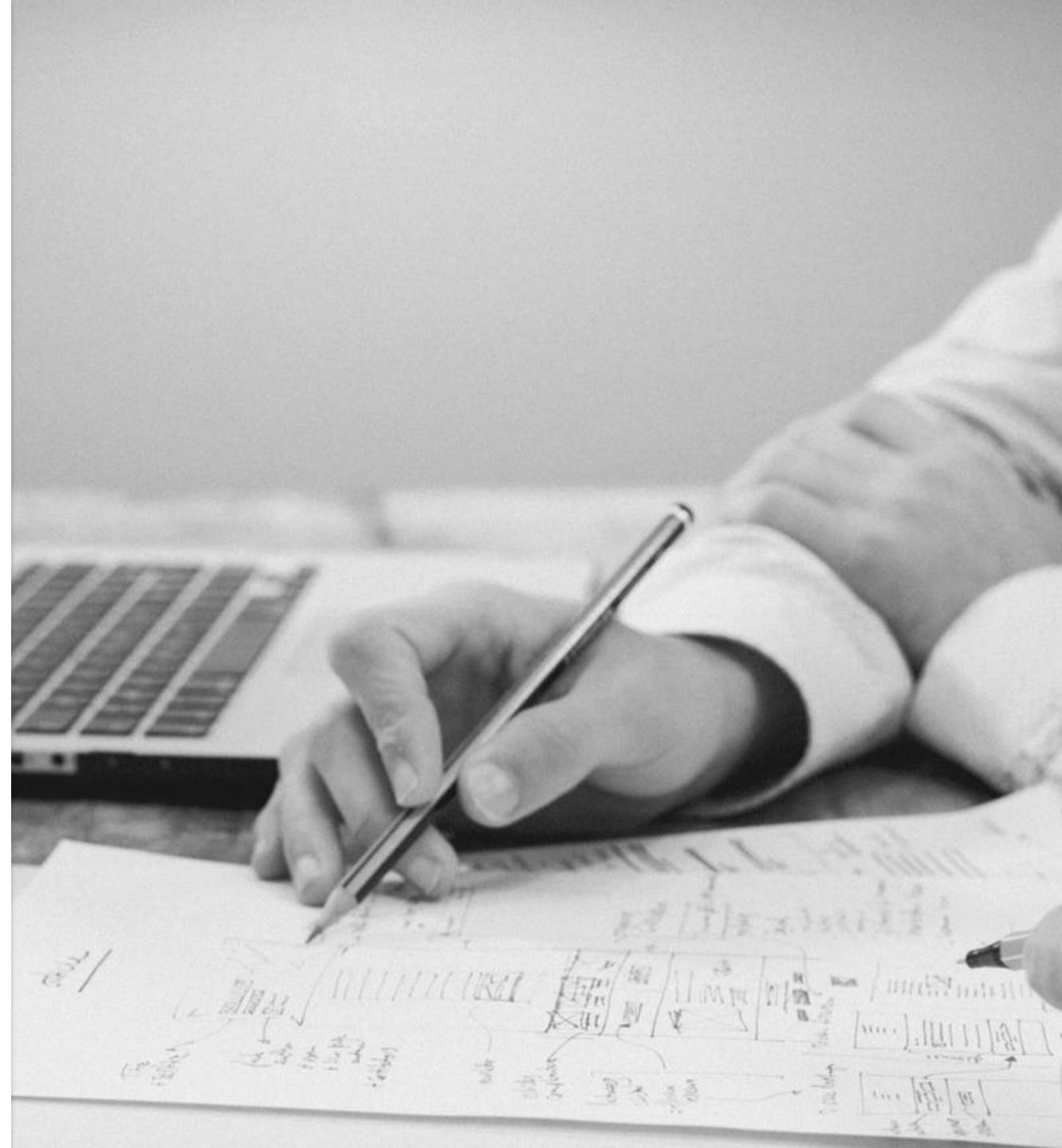
Administering the platform





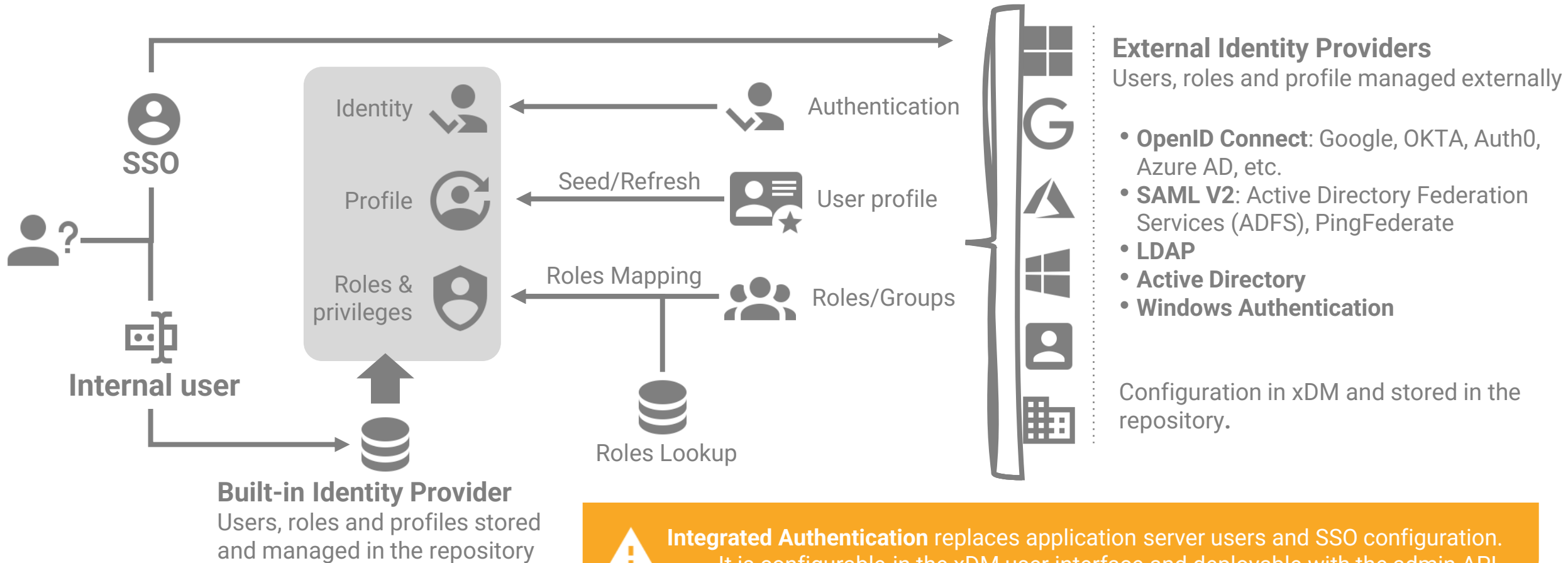
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Integrated Authentication



Integrated Authentication

Built-in Authentication and SSO Configuration



Identity Management

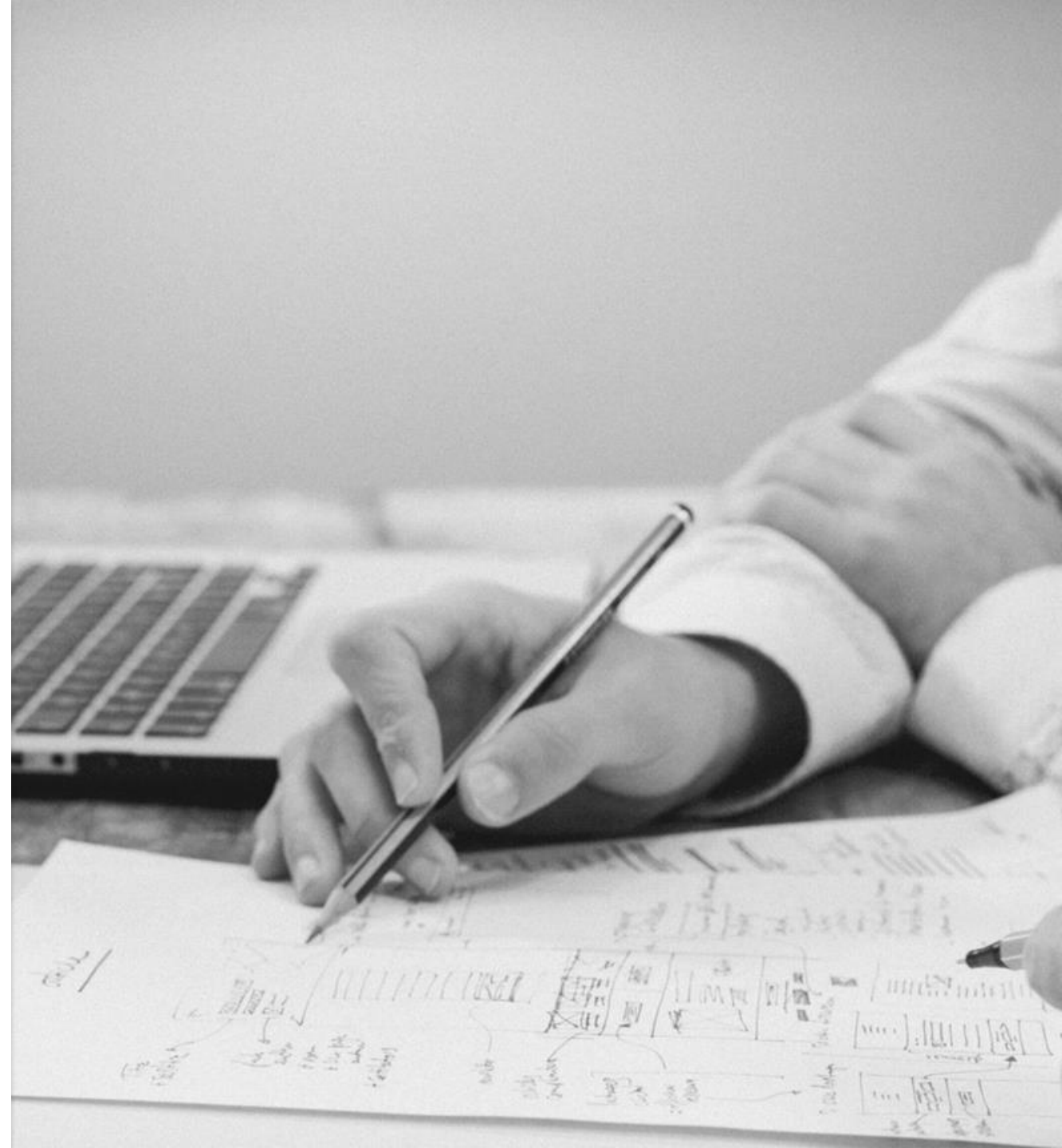
- Built-in configuration & testing
- Internal Identity Provider
- External IDP:
 - OpenID Connect
 - SAML
 - LDAP
 - Active Directory
 - Windows Authentication (SSO and Form)
- Roles & profile mapping
- Login page settings

The image displays two overlapping screenshots from the Semarchy xDM interface. The background screenshot shows the 'Google (OpenID Connect)' configuration page, which includes tabs for 'CONFIGURATION', 'ROLES MAPPING', 'PROFILE', and 'BUTTON'. The 'General' section contains fields for 'Name*' (Google), 'Provider type*' (OpenID Connect), and a 'Latest Connection' status showing a failure on April 7, 2021. The 'Configuration Properties' section includes 'Connectivity' details such as 'Issuer Identifier*' (https://accounts.google.com), 'Client ID*' (redacted), 'Client Secret*' (redacted), 'Redirect URL' (/login/oauth2/code/Google), and 'Logout URL' (https://accounts.google.com/logout). The 'User Authentication' section shows 'Additional Scopes' (email) and 'Username Claim*' (email). The foreground screenshot shows a login form with the Semarchy xDM logo, a 'LOGIN WITH GOOGLE' button, an 'OR' separator, and fields for 'User name' and 'Password', followed by a 'LOG IN' button.



Administering the platform

Integrated Datasources



Startup Configuration

- Startup Configuration required:
 - **Repository Datasource** configuration
 - Secrets Management configuration – Optional
- Properties set using one of the following methods:
 - Java System Properties
 - Environment Variables
 - Configuration File
- First startup of xDM application requires a **Setup Token** uses to authenticate and create the repository
 - Set as an environment variable:
SEMARCHY_SETUP_TOKEN=mySetupToken



Script samples are provided with the online [product documentation](#)

Repository Datasource

- Repository datasource is now configured with a **startup configuration mechanism**
- Startup configuration looks up the configuration properties in descending priority from:
 - **Java System Properties**
 - **Environment Variables**
 - **Configuration File**
- **A read-only user is required**

Oracle Sample - Configuration file

```
xdm.repository.url=jdbc:oracle:thin:@localhost:1521:ORA12
xdm.repository.driver=oracle.jdbc.OracleDriver
# Configure the repository datasource
xdm.repository.username=SEMARCHY_REPOSITORY
xdm.repository.password=SEMARCHY_REPOSITORY
# Configure the read-only datasource for dashboard
#xdm.repository.readonly.username=SEMARCHY_REPOSITORY_RO
#xdm.repository.readonly.password=SEMARCHY_REPOSITORY_RO
```

Platform Datasources

- **Datasources** are connections to databases storing data hubs, xDM Dashboard or xDM Discovery
- Declared from the **Platform Administration** or **REST API**
- **Datasource Usages** needs to be specified:
 - Data Locations
 - Dashboards
 - Discovery Sources
 - Database Reverse Engineering
 - Variable Value Providers
 - Enricher and Validator Plug-ins
- **Built-in test**

SEMARCHY_CUSTOMER_B2C_MDM

CONFIGURATION **ADVANCED**

General

Name: SEMARCHY_CUSTOMER_B2C_MDM

Description: Datasource created by upgrade process as a replacement of the jndi datasource java:comp/env/jdbc/SEMARCHY_CUSTOMER_B2C_MDM. Please complete the configuration.

URL*: jdbc:postgresql://localhost:15432/postgres

Driver*: org.postgresql.Driver

Username: semarchy_customer_b2c_mdm

Password: *****

Minimum Idle: 1

Maximum Pool Size: 8

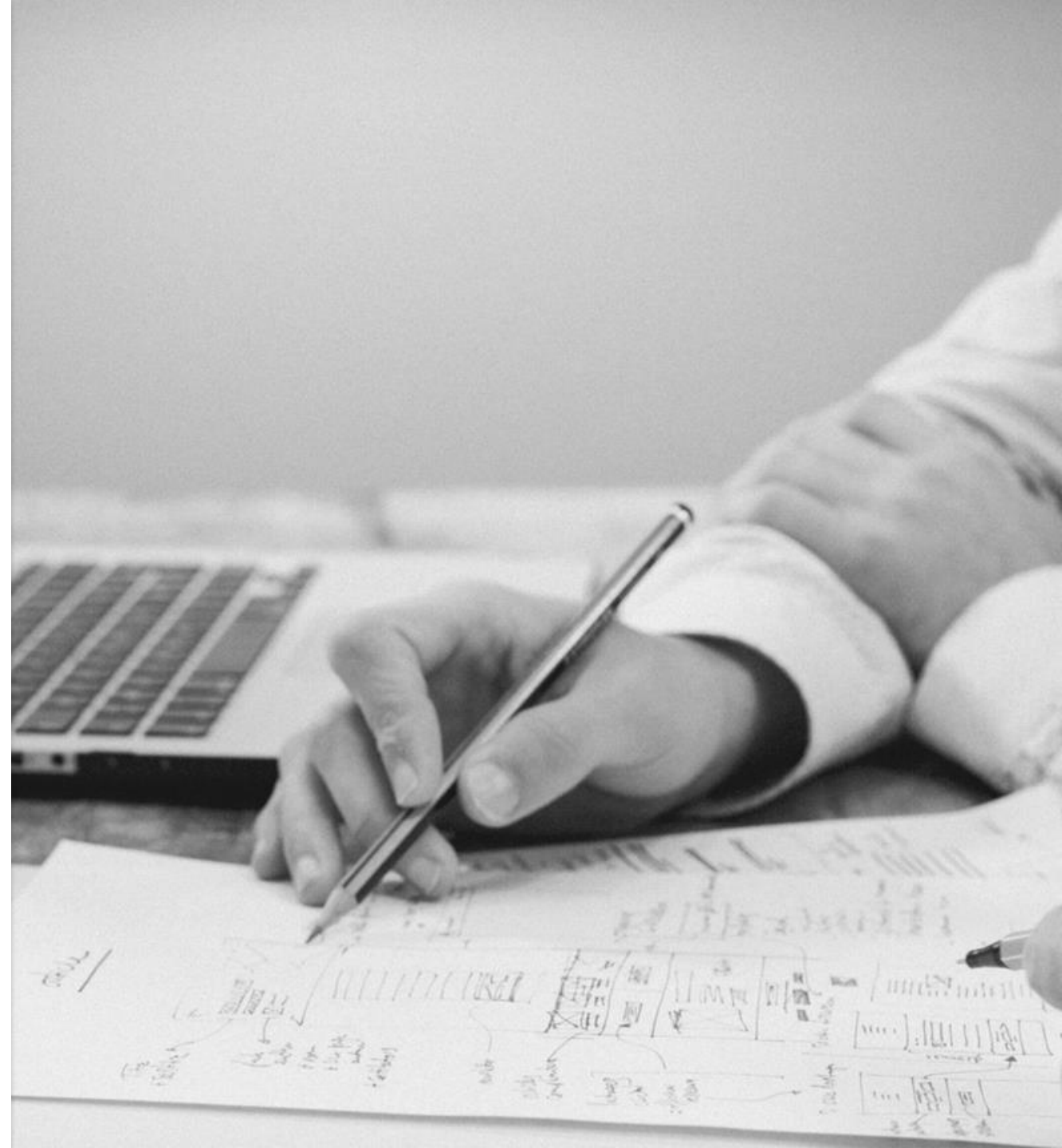
Datasource Usages

- ☒ **Data Locations**
Deploy models and store hub data in this datasource
- ☐ **Dashboards**
Use this datasource for xDM Dashboards queries
- ☐ **Discovery Sources**
Profile this datasource in xDM Discovery
- ☐ **Database Reverse Engineering**
Reverse-engineer models from tables in this datasource



Administering the platform

Secrets Management

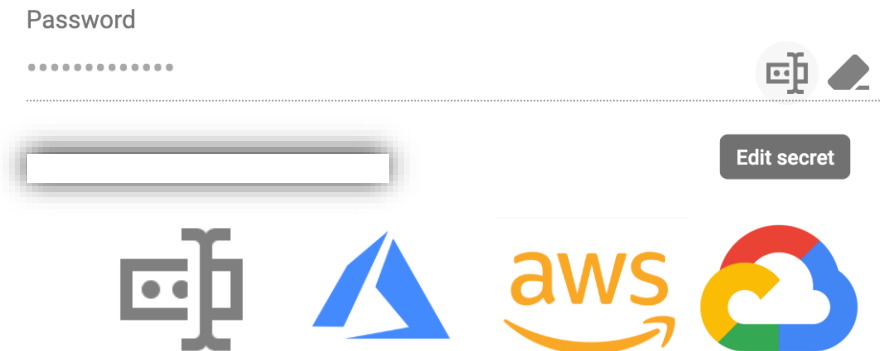


Secrets Management

Definition

Secret: Sensitive data that need to be encrypted and possibly decrypted.

- **Hashed Secrets:**
 - Stored and compared under hashed form
e.g.: User passwords, xDM API keys
- **Encrypted Secrets:**
 - Stored encrypted, decrypted when used.
E.g.: Database passwords, plugin parameters, AWS or Google API keys.
 - Stored in xDM, encrypted with a **local key** or using a third-party **Key Management Service**.
 - Stored in a third-party **Secrets Manager** and referenced from xDM



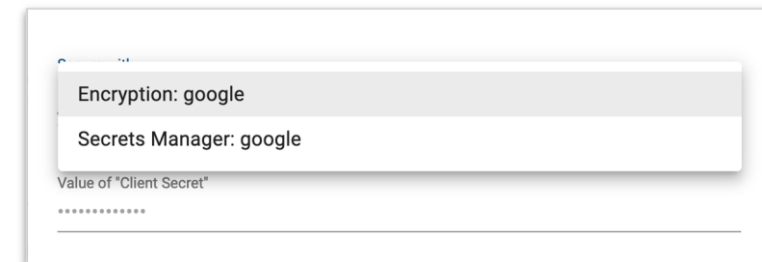
Key Management Service vs. Secrets Manager

Key Management Service (KMS)

- Provides an encryption key for xDM to encrypt and decrypt data stored locally in the repository
- One active KMS used to encrypt secrets
- Supported KMS
 - Local key
 - AWS KMS
 - Azure Key Vault
 - Google Cloud Key Management

Secrets Manager

- Stores encrypted secrets, which are *referenced* by xDM.
- Multiple active Secrets Managers
- Supported Secrets Managers
 - AWS Secrets Manager
 - Azure Key Vault
 - Google Cloud Secret Manager



Configuring Secrets in xDM

- Secrets are stored by default using a built-in encryption key (**insecure**)
- Secrets management configuration is loaded at startup **from properties**, looked up in descending priority from:
 - **Java System Properties**
E.g.: `-D=xdm.secrets.internal.kms.current=mykmsalias`
 - **Environment Variables**
Derived from the property name by upper casing and replacing "." by "_".
E.g.: `XDM_SECRETS_INTERNAL_KMS_CURRENT=mykmsalias`
 - A **Configuration File** containing properties
Default config file location: `<USER_HOME>/ .xdm/config.properties`

Properties Patterns

- Property names follow a pattern:
`xdm.secrets.internal.kms.mylocalkey.type=LOCAL`
`xdm.secrets.internal.kms.mylocalkey.keystore.file=~/.private/keystore.pkcs12`
`xdm.secrets.external.myaws.type=AWS`
`xdm.secrets.external.myaws.region=US-WEST-1`
- Pattern content
 - **Scope:** `internal.kms` for a KMS, `external` for a Secrets Manager
 - **Alias:** Alias of the KMS or Secrets Manager being configured (must be lowercase)
 - **Property:** Property to configure for the KMS or secrets manager.
- Special property: The current KMS
`xdm.secrets.internal.kms.current=mylocalkey`
- **IMPORTANT:** Property names should all be in lowercase (uppercased as ENV_VARIABLES)
- Configuration samples are available in the Preconfigured Bundle